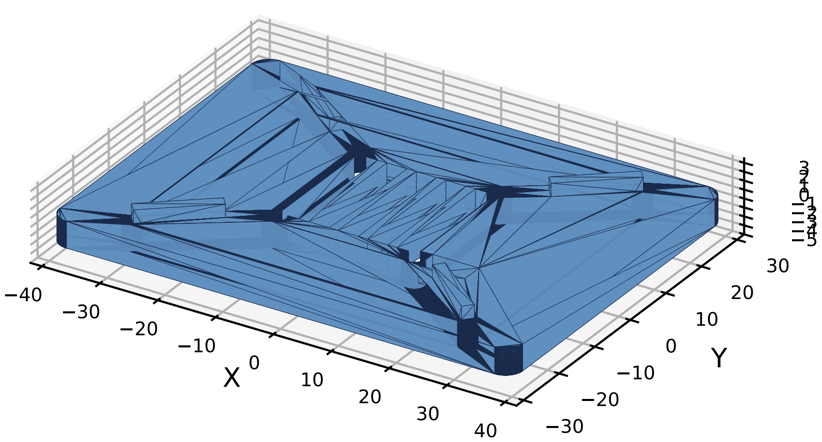
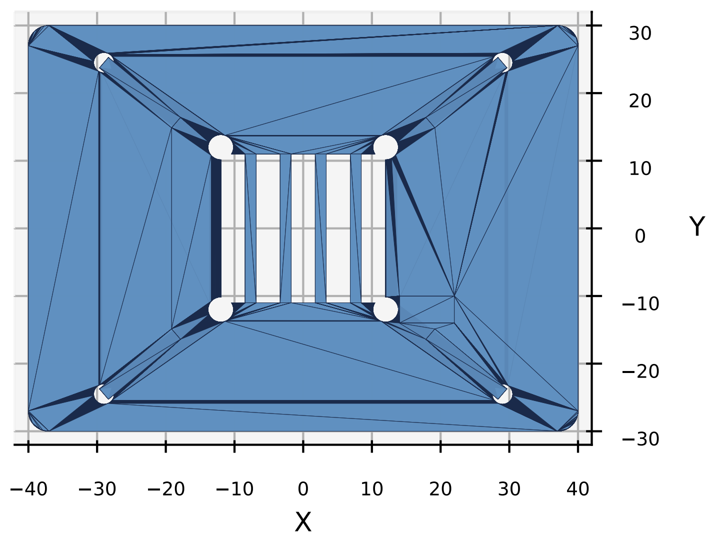
CSM V3 -- HMI Module 10 -- Pi Fan Cover V1.0

Active cooling for Pi 4: 30 x 30 x 10 mm fan, hangs below Pi PCB

Isometric View



Top View (fan side) -- 4x Pi + 4x fan + opening

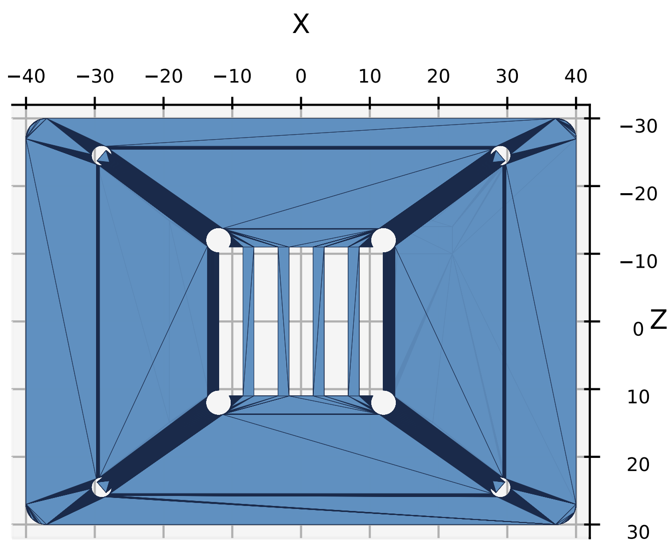


PART	Pi Fan Cover
NUMBER	M10-P04
VERSION	V1.2 (RC2)
MODULE	10 -- HMI
QTY	1
MATERIAL	PETG (PA12 prod)
INFILL	100%, 4 walls
LAYER	0.2 mm
MASS	~16.5 g
PRINT	flat, ~30 min

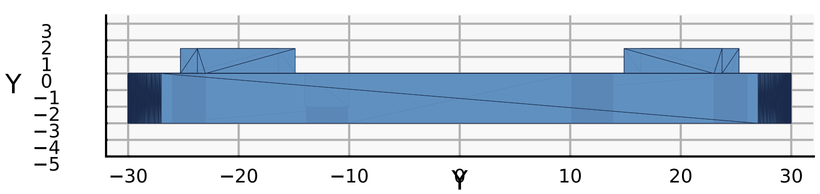
DIMENSIONS (mm)

Plate	80 x 60 x 3
Pi mount PCD	58 x 49 (Pi 4)
Pi screws	4x M2.5
Fan mount PCD	24 x 24 (30 mm fan)
Fan screws	4x M3
Fan grille	5 slots @ 3.5 x 22
Fan body	30 x 30 x 10
Pi-Cover gap	11 mm standoffs

Bottom View (operator side)



Side View -- 3 mm thickness



ASSEMBLY (after Pi Carrier mount)

1. Print PETG flat, 100% infill, 4 walls
2. Remove 4x M2.5 x 6 screws holding Pi to Pi Carrier standoffs (keep standoffs in)
3. Thread 4x 11 mm brass standoffs onto Pi bottom (sandwich Pi between carrier and these new standoffs)
4. Mount fan to Cover TOP face; fan AIRFLOW ARROW must POINT TOWARD Pi PCB
5. Align Cover under Pi; pass 4x M2.5 x 25 screws UP through Cover into standoffs
6. Wire fan: red->Pi pin 4 (+5V),
   black->Pi pin 6 (GND)

MOUNT STACK (top to bottom):
Pi Carrier (10 mm standoffs)
| Pi 4 PCB
| 11 mm brass standoffs
| Pi Fan Cover (3 mm)
| M2.5 x 25 screw heads

WALL CHECK: all 7 OK, min 4.0 mm